

Calculating the determinant using Kraut method in $O(N^3)$

In this article, we'll describe how to find the determinant of the matrix using Kraut method, which works in $O(N^3)$.

The Kraut algorithm finds decomposition of matrix A as $A = LU$ where L is lower triangular and U is upper triangular matrix. Without loss of generality, we can assume that all the diagonal elements of L are equal to 1. Once we know these matrices, it is easy to calculate the determinant of A : it is equal to the product of all the elements on the main diagonal of the matrix U .

There is a theorem stating that any invertible matrix has a LU-decomposition, and it is unique, if and only if all its principle minors are non-zero. We consider only such decomposition in which the diagonal of matrix L consists of ones.

Let A be the matrix and N - its size. We will find the elements of the matrices L and U using the following steps:

1. Let $L_{ii} = 1$ for $i = 1, 2, \dots, N$.
2. For each $j = 1, 2, \dots, N$ perform:
 - For $i = 1, 2, \dots, j$ find values

$$U_{ij} = A_{ij} - \sum_{k=1}^{i-1} L_{ik} \cdot U_{kj}$$
 - Next, for $i = j+1, j+2, \dots, N$ find values

$$L_{ij} = \frac{U_{ij}}{U_{jj}} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik} \cdot U_{kj} \right)$$

Implementation

```
static BigInteger det (BigDecimal a [][], int n) {
    try {

        for (int i=0; i<n; i++) {
            boolean nonzero = false;
            for (int j=0; j<n; j++)
                if (a[i][j].compareTo (new BigDecimal (BigInteger.ZERO)) > 0)
                    nonzero = true;
            if (!nonzero)
```